Nikolay Dimitrov

Automotive Systems Engineer

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Summary

Dynamic and seasoned **Automotive Systems Engineer** with over 8 years of experience, specializing in **system architecture, specification, and embedded software development** within the automotive industry. Proven track record designing and developing efficient, robust, and safety-critical systems using **AUTOSAR methodology**. Deep understanding of **system architecture principles**, requirements analysis, microcontroller programming, and system integration, gained from successfully leading complex projects (including **ASIL-D/B**) in fast-paced environments. Familiarity with automotive standards like **ASPICE**. Passionate about innovative technology solutions for **digital cockpits and electrification**. Committed to driving system-level solutions and maintaining high quality standards.

Skills Summary

System Design & Architecture

- System Architecture Design & Specification
- Requirements Analysis & Management (IBM Doors)
- System Integration Strategy & Planning
- Functional Safety (ISO 26262, ASIL D/B Concepts)
- DFMEA Principles (Conceptual Understanding)
- Multi-core System Design (IPC, Sync, Partitioning)
- Real-Time Systems Design & Timing Analysis

Automotive & Embedded

- AUTOSAR Methodology (Classic Platform: BswM, EcuM, RTE, Configuration, Integration, .arxml)
- Embedded C Programming
- Microcontroller Programming (ARM Architecture)
- ECU Boot-up & Shutdown Sequences
- Advanced Driver Assistance Systems (ADAS Concepts)

Communication & Protocols

- CAN Bus
- Inter-Process Communication (IPC: Shared Memory, Message Queues, RPC)
- End-to-End (E2E) Protection Concepts

Tools & Technologies

- Vector Toolchain (DaVinci Configurator/Developer, CANoe, Candela)
- Architecture/Requirements Tools (IBM Rhapsody, IBM Doors)
- Build Systems (CMake, Jenkins)
- Debugging Tools (Multi-core debugging experience)

Methodologies & Standards

- ASPICE (Familiarity)
- V-Model Development Process

Professional Experience

Sensified

02/2022 – Present, Bulgaria, Sofia

At Sensified, we form a dynamic and highly professional team committed to embedded engineering consultancy. As a consultant, I thrive on adapting to diverse client requirements and consistently delivering results within tight deadlines. Currently, I am deeply engaged in two ASIL projects at Sensified: one categorized as ASIL-D and the other as ASIL-B, focusing on system safety and architecture.

System Engineer - External consultant (Sensified), Continental

02/2022 – 11/2024, Germany, Ulm

Integral member of the Advanced Driver Assistance Systems (ADAS) division, focused on **System Architecture and Specification** for safety-critical components, including autopilot and parking assistance functionalities (ASIL D/B).

- **Analyzed customer specifications** and performance requirements to define system-level design expectations for complex ADAS features.
- Developed **System Architecture**, detailed **System Specifications**, and **Technical Concepts** for the safety core functionalities deployed on a complex multi-core microcontroller.
- **Led technical clarifications** and defined solutions through collaboration with cross-functional engineering teams (Software, Hardware) and the Customer, ensuring alignment on system design and integration.
- Defined the **Systems Integration strategy** for the multi-core safety platform, including boot-up, shutdown, and inter-core communication approaches.
- Specified robust ECU boot-up and shutdown behaviors within the **system architecture**, leveraging AUTOSAR methodologies (BswM, EcuM) and ensuring correct component configuration from HSM to Application Software.
- Architected and specified system-level **Inter-Process Communication (IPC)** mechanisms (shared memory, message queues, RPC) based on analysis of inter-core communication requirements.
- Specified **synchronization strategies** (semaphores, mutexes, spinlocks) as part of the system design to safeguard shared resources and ensure system stability in concurrent multi-core environments.
- Defined **partitioning strategies** (spatial and temporal) within the system architecture to isolate critical functions, directly addressing system safety (ASIL) and security requirements.
- Conducted **system-level timing analyses** and specified scheduling strategies to ensure the system met deterministic real-time performance targets.
- Supported system integration and validation phases by performing **analysis** of multi-core behavior and driving **problem-solving** for complex system-level issues.
- *(Optional - Verify Accuracy)* Contributed to **System DFMEA** activities related to the safety core functionalities.
- *(Optional - Verify Accuracy)* Utilized **requirements management** (IBM Doors) and **architecture design tools** (IBM Rhapsody) during the system definition phase.

Software Engineer - Internal Employee, BHTC

02/2017 – 02/2022, Bulgaria, Sofia

Key contributor within the AUTOSAR and SW-PP divisions, developing and integrating standard software components for multiple projects. Focused on basic software development, project initiation according to BHTC-specific requirements, and configuration based on system/customer needs.

- Integrated, configured, and tested AUTOSAR, non-AUTOSAR, and third-party software modules according to customer requirements and hardware specifications.
- Utilized key development tools including IBM Doors (requirements), IBM Rhapsody (design), Vector DaVinci Suite (Configurator/Developer), CANoe, and Candela.
- Involved in the full project lifecycle, from initiation and component design through to testing and documentation across multiple projects.
- Mentored new team members, fostering knowledge sharing and a collaborative environment.
- Provided cross-functional support to colleagues across global BHTC locations, aiding in component integration and system understanding.

Junior Embedded Software Engineer, Ansisoft Ltd.

01/2015 - 02/2017, Bulgaria, Sofia

Executed application development for the open-source electronic prototyping platform, Arduino.

- Worked with C programming on ARM Architecture.
- Developed proof-of-concept applications using C#.

CEO and Co-founder, MMA Bulgaria

2011 - 2015, Bulgaria

Founded mma.bg, a hub for Mixed Martial Arts enthusiasts providing news, event updates, training resources, and community engagement.

- Led strategic decision-making and managed a startup from inception to growth, overseeing operations, logistics, software development, and event management across multiple countries.

Education

Attempted PhD in Computer Technology in Engineering

2011 – Present, Bulgaria, Sofia

University of Mining and Geology

Master Degree in Information Technology

2010 – 2011, Bulgaria, Sofia

University of Mining and Geology

Bachelor in Automation, Information and Computer Science

2006 – 2009, Bulgaria, Sofia

University of Mining and Geology